Solved selected problems of Modern Physics for Scientists and Engineers - John Taylor

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Chapter 10 - Electron Spin

Solution. 10.1 Given that the charge density is given by $\rho(r) = -e|\psi(r)|^2$, and that the probability of finding the electron between a_B and ∞ is 0.67 then the enclosed charge is

$$Q = \int_0^{a_B} \rho(r)\ dr = -e \int_0^{a_B} |\psi(r)|^2\ dr = -e(1 - 0.67) = -0.33e$$

Using Gauss' law, taking a spherical Gaussian surface we see that

$$\begin{aligned}
\mathcal{E} \cdot 4\pi a_B^2 &= \frac{Q}{\varepsilon_0} \\
\mathcal{E} &= \frac{-0.33e}{4\pi r^2 \varepsilon_0} \\
\mathcal{E} &= \frac{0.33 \cdot 1.602 \times 10^{-19}}{4\pi \cdot (5.29 \times 10^{-11})^2 \cdot 8.854 \times 10^{-12}} \\
\mathcal{E} &= 169.79 \times 10^9\ V/m
\end{aligned}$$

□

Solution. 10.2

(a) Given that $\rho(r) = \rho_0 e^{-r/R}$ is the average charge distribution of $Z-1$ electrons then must be that

$$\int \rho(r)\ dV = Q$$

$$4\pi \int_0^\infty \rho_0 e^{-r/R} r^2\ dr = -(Z-1)e$$

$$4\pi\rho_0[2R^3] = -(Z-1)e$$

$$\rho_0 = \frac{-(Z-1)e}{8\pi R^3}$$

(b) Let us compute first the charge Q enclosed at a radius r

$$\begin{aligned} Q &= Ze + 4\pi \int_0^r \rho(r) r^2\ dr \\ &= Ze - \frac{(Z-1)e}{2R^3} \int_0^r e^{-r/R} r^2\ dr \\ &= Ze - \frac{(Z-1)e}{2R^3} \left[2R^3 - Re^{-r/R}(r^2 + 2rR + 2R^2)\right] \\ &= e + \frac{(Z-1)e}{2R^2} e^{-r/R}(r^2 + 2rR + 2R^2) \end{aligned}$$

Now, taking a spherical Gaussian surface we see that

$$\begin{aligned} \mathcal{E} \cdot 4\pi r^2 &= \frac{Q}{\varepsilon_0} \\ \mathcal{E} &= \frac{1}{4\pi\varepsilon_0 r^2}\left[e + \frac{(Z-1)e}{2R^2} e^{-r/R}(r^2 + 2rR + 2R^2)\right] \\ \mathcal{E} &= \frac{k}{r^2}\left[e + \frac{(Z-1)e}{2R^2} e^{-r/R}(r^2 + 2rR + 2R^2)\right] \end{aligned}$$

(c) We see that as $r \to 0$ then $Q \to Ze$ so

$$\mathcal{E} = \frac{Zke}{r^2}$$

Which matches with the result of equation (10.2).
If $r \to \infty$ then $Q \to e$ so

$$\mathcal{E} = \frac{ke}{r^2}$$

Which matches with the result of equation (10.3) as we wanted.

□

Solution. 10.3

(a) The innermost electron of lead (1s) has $Z_{\text{eff}} \approx Z$ and the wave function approximates that of a hydrogen-like ion with nuclear charge Ze and hence the 1s energy is given by

$$E_{1s} = -Z^2 E_R = -(82)^2 \cdot 13.6\ eV = -91446.4\ eV$$

(b) The most probable radius of this electron is a_B/Z so

$$\frac{a_B}{Z} = \frac{0.0529}{82}\ nm = 0.0006451\ nm$$

$\square$

Solution. 10.5

(a) Assuming the outermost electron is far enough to the rest of the electrons the outer electron would feel a potential-energy as one electron in hydrogen, i.e.

$$U(r) = -\frac{ke^2}{r}$$

(b) We can say that an electron in a $3d$ state is moving outside an effective charge of $+e$, so we may approximate the energy of such an electron as

$$E_{3d} = -\frac{E_R}{Z^2} = -\frac{13.6}{11^2} = -0.112\ eV$$

Our approximation is a lot smaller than the measured value of $-1.52\ eV$. A plausible explanation is that our assumption of the outermost electron behaving like an individual electron, as in the hydrogen atom, is wrong, and there is more interaction between the $3d$ electron and the rest of the electrons, or in other words, the innermost electrons are not far enough to take this approximation.

$\square$

Solution. 10.6

(a) Assuming the outermost electron is far enough to the rest of the electrons the outer electron would feel a potential-energy as the only electron in the hydrogen atom, i.e.

$$U(r) = -\frac{ke^2}{r}$$

(b) We can say that an electron in the outermost shell is moving outside an effective charge of $+e$, so $Z_{\text{eff}} \approx 1$ we may approximate the energy of such an electron as

$$E_n = -\frac{E_R}{n^2}$$

(c) The energy of a $3p$ electron can be approximated as

$$E_{3p} = -\frac{E_R}{3^2} = -\frac{13.6}{3^2} = -1.511\ eV$$

This approximation works well compared to the observed value of $1.556\ eV$.

(d) In the case that the outer electron is in the $3d$ level we get as well that $E_{3d} = -1.511\ eV$ which is even closer to the observed value of $-1.513\ eV$.

(e) The approximation is better for the $3d$ level because the wave function for this electron is not as penetrating as the wavefunction for the $3p$ electron, so more of the electron is outside of the $3s$ electrons, so the approximation of this electron being attracted by an effective charge of $+e$ is better for the $3d$ electron than for the $3p$ electron.

The observed energy for a $3p$ electron is lower than for the $3d$ electron because it's closer to the nucleus, so to remove it we need to put more energy.

□

Solution. 10.7

(a) If $l = 2$ then m can take the values $2, 1, 0, -1$, or -2 and s can be $1/2$ or $-1/2$ so we can acomodate $5 \times 2 = 10$ electrons.

(b) If $l = 3$ then m can take the values $3, 2, 1, 0, -1, -2$, or -3 and s can be $1/2$ or $-1/2$ so we can acomodate $7 \times 2 = 14$ electrons.

(c) We note that if $l = 0$ we can acomodate 2×1 electrons (with different spins), if $l = 1$ we can acomodate 2×3 electrons, if $l = 2$ we can acomodate 2×5 electrons, and so on. Thenthe number of electrons we can acomodate can be put as

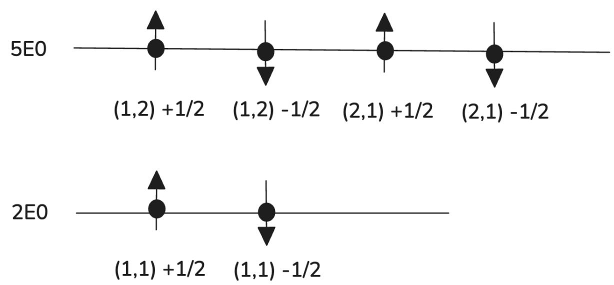
$$\text{nr of electrons} = 2(2l + 1)$$

$\square$

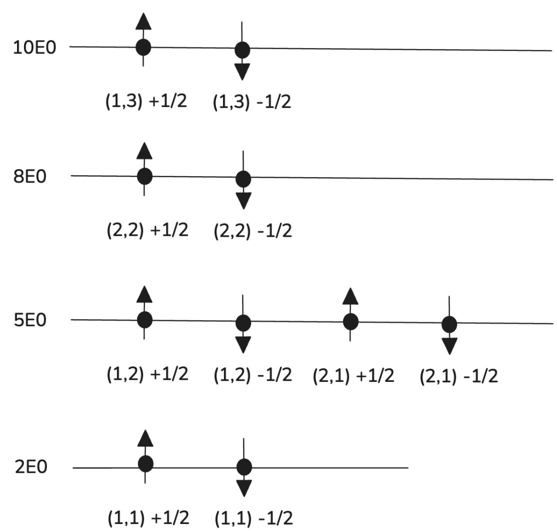
Solution. 10.11

(a) The lowest four allowed energy levels for each particle are $2E_0, 5E_0, 8E_0$, and $10E_0$.
The energy level $2E_0$ can accomodate 2 particles since n_x and n_y can only take the value 1 but the spin can be $\pm 1/2$.
The energy level $5E_0$ can accomodate 4 particles since n_x and n_y can take the values 1 or 2 (2 options) and the spin can be $\pm 1/2$.
The energy level $8E_0$ can accomodate 2 particles since n_x and n_y can only take the value 2 and the spin can be $\pm 1/2$.
The energy level $10E_0$ can accomodate 4 particles since n_x and n_y can take the values 1 and 3 and the spin can be $\pm 1/2$.

(b) We can accomodate 6 particles in the box as follows



(c) We can accomodate 10 particles in the box as follows



$\square$

Solution. 10.12

(a) Let us consider the wave function $\psi(x_1, x_2) = e^{-(x_1^2+x_2^2)}$, we note that

$$\psi(x_1, x_2) = e^{-(x_1^2+x_2^2)} = e^{-(x_2^2+x_1^2)} = \psi(x_2, x_1)$$

So the wave function is symmetric. Also, we see that

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |\psi(x_1, x_2)|^2 \; dx_1 dx_2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-2(x_1^2+x_2^2)} \; dx_1 dx_2 = \pi/2$$

which implies that the wave function is normalizable.

(b) Let us consider the wave function $\psi(x_1, x_2) = x_1 - x_2$, we note that

$$\psi(x_1, x_2) = x_1 - x_2 = -(x_2 - x_1) = -\psi(x_2, x_1)$$

So the wave function is antisymmetric.

(c) Let us consider the wave function $\psi(x_1, x_2) = 2x_1 - x_2$, we note that

$$\psi(x_1, x_2) = 2x_1 - x_2 = -(x_2 - 2x_1) \neq -(2x_2 - x_1) = -\psi(x_2, x_1)$$

So the wave function is not antisymmetric, also, let us note that

$$\psi(x_1, x_2) = 2x_1 - x_2 \neq 2x_2 - x_1 = \psi(x_2, x_1)$$

So the wave function is neither symmetric.

□

Solution. 10.13

(a) We know that electrons are fermions so if we have an electron in the state $(1s, 1/2)$ then the other must be in the state $(1s, -1/2)$, and since they are indistinguishable swapping them is not considered as a new atom state, so there is just one independent state of the whole atom and hence the degeneracy is 1.

(b) Suppose now that instead of having electrons we have half-spin bosons then the possible states for the bosons are

$$(1s, 1/2)\ (1s, 1/2) \quad (1s, -1/2)\ (1s, -1/2) \quad (1s, 1/2)\ (1s, -1/2)$$

Again, since bosons are indistinguishable swapping the last state is not considered an independent state of the atom, therefore the degeneracy is 3.

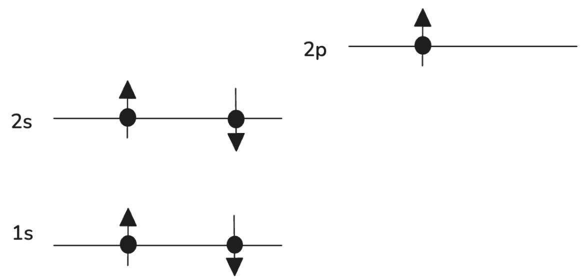
(c) If somehow the two electrons were distinguishable then the possible states are

$$\begin{array}{cc} (1s, 1/2)\ (1s, 1/2) & (1s, -1/2)\ (1s, -1/2) \\ (1s, 1/2)\ (1s, -1/2) & (1s, -1/2)\ (1s, 1/2) \end{array}$$

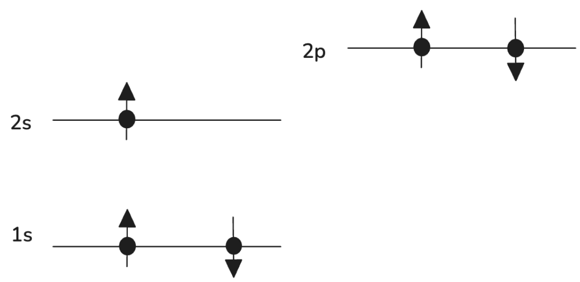
In this case, we can have the same spin for the two elecrons since they can be distinguished by their mass. Therefore the degeneracy in this case is 4.

□

Solution. 10.14 For Boron $_5B$ the energy-level diagram for the ground state looks like



And for the first excited state looks like the following



$\square$

Solution. 10.17
When jumping from ${}_4Be$ to ${}_5B$ we have two competing effects: The increase in Z causes any given level to be somewhat more tightly bound, but the final electron has to occupy a level that is slightly higher and hence somewhat less tightly bound.
For the ionization energy, the second effect wins, so the ionization energy for ${}_5B$ is slightly lower than the one for ${}_4Be$. $\square$

Solution. 10.19

(a) Treating each electron separately as if it were in a hydrogen-like ion we see that each one has an energy of

$$E = -Z^2 E_R = -4 \cdot 13.6 \ eV = -54.4 \ eV$$

Therefore in this approximation the total energy of a helium atom is

$$E_{He} = 2 \cdot -54.4 \ eV = -108.8 \ eV$$

(b) Assuming both electrons are in the first Bohr orbit with radius $a_B/2$ on opposite sides of the atom, separated a distance a_B then the potential energy between the two electrons is

$$\begin{aligned} U &= \frac{ke^2}{a_B} \\ &= \frac{8.9877 \times 10^9 \cdot (1.602 \times 10^{-19})^2}{5.29 \times 10^{-11}} \\ &= 4.3603 \times 10^{-18} \ J = 27.214 \ eV \end{aligned}$$

Therefore the total energy of a helium atom adding the potential energy between the electrons is

$$E_{He} = -108.8 + 27.214 = -81.586 \ eV$$

This is a close approximation to the measured value of $-79.0 \ eV$.

$\square$

Solution. 10.23
We see that Cd ($Z = 48$) has it's $4d$ level full and hence In ($Z = 49$) has 1 electron at the level $5p$, so In has an electron in a level with higher energy and hence it's less tightly bound, so the ionization energy has a drop.
In the same way, Hg ($Z = 80$) has from the 6th shell the levels $4f$, $5d$ and $6s$ full of electrons. They are filled in that order since, for example, $4f$ has lower energy than $5d$. Then Tl ($Z = 81$) has 1 electron at the next sub-shell $6p$ which has higher energy and hence is less tightly bound, therefore the ionization energy has a drop. $\square$

Solution. 10.27

If the electron had a spin $s = 3/2$ then there are 4 possible spins $-3/2, -1/2, 1/2$ and $3/2$ so the s level can have 4 electrons, the p level can have 12 electrons and the d level can have 20 electrons.

Then the ground-state configurations for ${}_8O$, ${}_{10}Ne$ and ${}_{21}Sc$ are

$$\begin{aligned} {}_8O: &\quad 1s^4 2s^4 \\ {}_{10}Ne: &\quad 1s^4 2s^4 2p^2 \\ {}_{21}Sc: &\quad 1s^4 2s^4 2p^{12} 3s^1 \end{aligned}$$

□

Solution. 10.33

(a) Assuming the ionization energies vary linearly from ${}_{20}Ca$ to ${}_{38}Sr$ then the slope of the linear interpolation formula is

$$m = \frac{5.70 - 6.11}{38 - 20} = -0.0227$$

So for ${}_{56}Ba$ we have that

$$y = y_1 + m(x - x_1) = 6.11 - 0.0227(56 - 20) = 5.292\ eV$$

This value is close to the observed value of 5.21 eV.

(b) Assuming the electron affinity vary linearly from ${}_{17}Cl$ to ${}_{53}I$ then the slope of the linear interpolation formula is

$$m = \frac{3.06 - 3.61}{53 - 17} = -0.01527$$

So for ${}_{35}Br$ we have that

$$y = y_1 + m(x - x_1) = 3.61 - 0.01527(35 - 17) = 3.335\ eV$$

(c) Assuming the atomic radius vary linearly from ${}_{7}N$ to ${}_{8}O$ then the slope of the linear interpolation formula is

$$m = \frac{0.065 - 0.075}{8 - 7} = -0.01$$

So for ${}_{9}F$ we have that

$$y = y_1 + m(x - x_1) = 0.075 - 0.01(9 - 7) = 0.055\ nm$$

$\square$

Solution. 10.35
Assuming the electron affinity vary linearly from ${}_{17}Cl$ to ${}_{35}Br$ then the slope of the linear interpolation formula is

$$m = \frac{3.36 - 3.62}{35 - 17} = -0.014$$

So for ${}_{9}F$ we have that

$$y = y_1 + m(x - x_1) = 3.62 - 0.014(9 - 17) = 3.73\ eV$$

Which is not that close to the observed value of 3.40 eV. □

Solution. 10.36
The energy of the $1s$ electrons is approximately $Z^2 E_R = -122.4\ eV$ for the lithium atom. Comparing this value to the ones in figure 10.16 we see that the lowest five excited levels must be

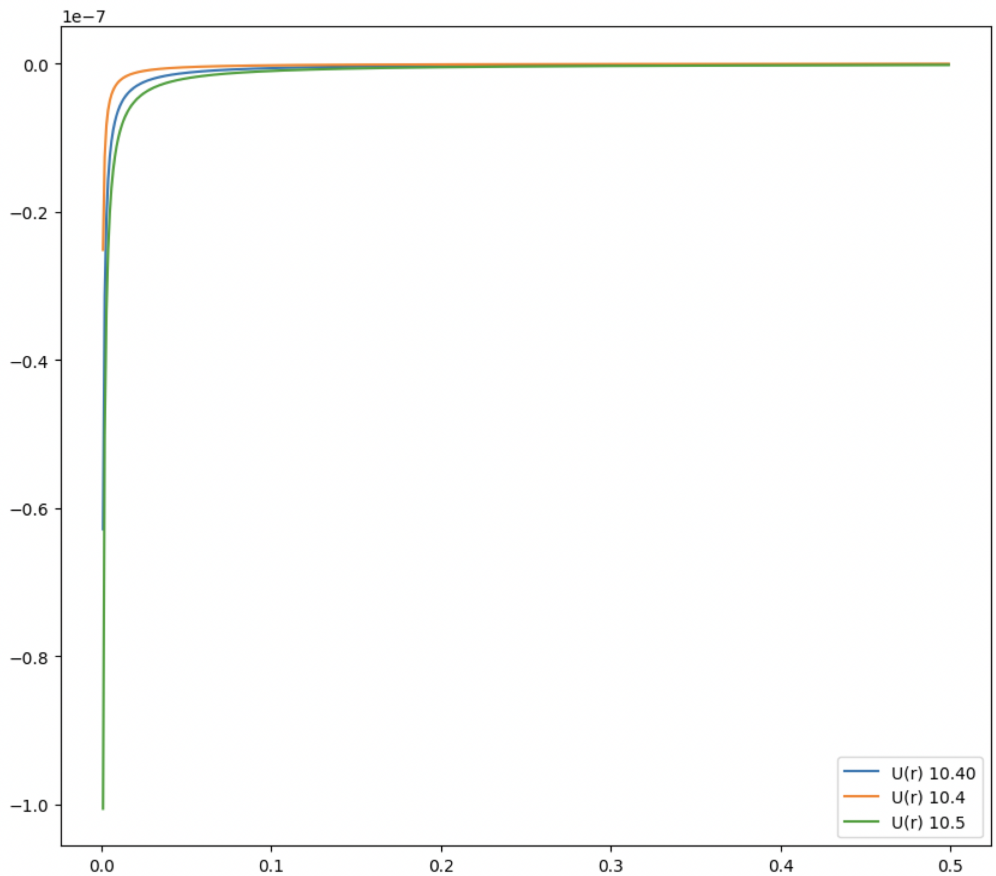
$$1s^2 2p^1 \qquad 1s^2 3s^1 \qquad 1s^2 3p^1 \qquad 1s^2 3d^1 \qquad 1s^2 4s^1$$

Since it requires less energy to move the $2s$ electron to a higher level compared to the $1s$ electrons. $\square$

Solution. 10.40
The potential energy $U(r)$ associated to $\mathcal{E}$ is given by

$$
\begin{aligned}
U(r) &= -\int (-e)\mathcal{E}\ dr \\
&= e^2k\int \frac{1}{r^2}\left[1+\frac{(Z-1)}{2R^2}e^{-r/R}(r^2+2rR+2R^2)\right]\ dr \\
&= e^2k\int \frac{1}{r^2}+\frac{(Z-1)}{2R^2}\left(e^{-r/R}+\frac{2Re^{-r/R}}{r}+\frac{2R^2e^{-r/R}}{r^2}\right)\ dr \\
&= e^2k\int \frac{1}{r^2}+\frac{(Z-1)}{2R^2}\left(e^{-r/R}+\frac{2Re^{-r/R}}{r}+\frac{2R^2e^{-r/R}}{r^2}\right)\ dr \\
&= e^2k\left[-\frac{1}{r}+\frac{(Z-1)}{2R^2}\left(-Re^{-r/R}+2R\mathrm{Ei}(-r/R)-2R\mathrm{Ei}(-r/R)-\frac{e^{-r/R}}{r}\right)\right] \\
&= -e^2k\left[\frac{1}{r}+\frac{(Z-1)}{2R^2}e^{-r/R}\left(R+\frac{1}{r}\right)\right]
\end{aligned}
$$

Ploting this equation and equations (10.4) and (10.5) we see that



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